

3. IoT-Equipped and AI-Enabled Next Generation Smart Agriculture: A Critical Review, Current Challenges and Future Trends

Why it matches your project

This paper directly discusses the integration of **Artificial Intelligence (AI)** and the **Internet of Things (IoT)** in smart agriculture. It covers intelligent irrigation, automated monitoring, and AI-based decision-making, which are key components of your smart farming robot.

Summary

The paper reviews next-generation smart agriculture systems that combine IoT devices, wireless sensor networks, cloud computing, edge computing, robotics, and artificial intelligence. It highlights applications such as precision irrigation, crop monitoring, disease prediction, and automated farming. The study emphasizes that AI and IoT technologies improve productivity while reducing water consumption and labor requirements. It also discusses challenges including cybersecurity, scalability, and connectivity issues. The authors conclude that AI-enabled IoT systems will play a crucial role in the future development of autonomous agriculture.